

Linear Algebra

Matrices

- A matrix is a rectangular array of numbers arranged in rows and columns.
- **Order:** Denoted as $m \times n$, where m is the number of rows and n is the number of columns.

This Matrix $[M]$ has 3 rows and 2 columns i.e., order of 3×2 . Each element of matrix $[M]$ can be referred to by its row and column number. For example, $a_{32} = 0$.

Types of Matrices

- **Square Matrix:** $m = n$.
- **Diagonal Matrix:** All non-diagonal elements are zero.
- **Identity Matrix (I):** A diagonal matrix with all diagonal elements as 1.
- **Zero Matrix:** All elements are zero.
- **Symmetric Matrix:** $A = A^T$ (Transpose equals the original matrix).
- **Skew-Symmetric Matrix:** $A = -A^T$.
- **Orthogonal Matrix:** $A^T A = I$, where A^T is the transpose.
- **Singular Matrix:** Determinant is 0 ($\det(A) = 0$).
- **Non-Singular Matrix:** Determinant is non-zero ($\det(A) \neq 0$).
- **Idempotent Matrix:** A matrix is said to be idempotent if $A^2 = A$.
- **Involutory Matrix:** A matrix is said to be Involutory if $A^2 = I$.
- **Nilpotent Matrix:** A square matrix of order n is said to be nilpotent if $A^k = 0$, $k \leq n$.

Operations on Matrices

Matrix Operations are basic calculations performed on matrices to solve problems or manipulate their structure.

- **Matrix Addition**
- **Matrix Multiplication**
- **Transpose of a Matrix**
- **Inverse of Matrix**

Transpose of a Matrix: The transpose $[M]^T$ of an $m \times n$ matrix $[M]$ is the $n \times m$ matrix obtained by interchanging the rows and columns of $[M]$. if $A = [a_{ij}]_{m \times n}$, then $A^T = [b_{ij}]_{n \times m}$ where $b_{ij} = a_{ji}$

Properties of transpose of a Matrix:

- $(A^T)^T = A$
- $(kA)^T = k(A^T)$
- $(A \pm B)^T = A^T \pm B^T$
- $(AB)^T = B^T \cdot A^T$
- $(A^{-1})^T = (A^T)^{-1}$

Adjoint of a Matrix: The adjoint of a square matrix is the transpose of its cofactor matrix.

Properties of Adjoint

Some important properties of adjoint include:

- $A(\text{Adj } A) = (\text{Adj } A) A = |A| I_n$
- $\text{Adj}(AB) = (\text{Adj } B) \cdot (\text{Adj } A)$
- $|\text{Adj } A| = |A|^{n-1}$
- $\text{adj}(\text{adj}(A)) = |A|^{n-2} \cdot A$
- $\text{adj}(A^T) = (\text{adj}(A))^T$
- $\text{Adj}(kA) = k^{n-1} \text{Adj}(A)$

Inverse of a Matrix

For any square matrix A,

$$A^{-1} = \frac{\text{Adj } A}{|A|}$$

Here $|A|$ should not be equal to zero, means matrix A should be non-singular.

Properties of Inverse

- $A^{-1} \cdot A = I$
- $(A^{-1})^{-1} = A$
- $(AB)^{-1} = B^{-1}A^{-1}$
- $(kA)^{-1} = (1/k) \cdot A^{-1}$

Note: Only a non-singular square matrix can have an inverse.

Trace of a Matrix: Let $A = [a_{ij}]_{n \times n}$ is a square matrix of order n, then the sum of diagonal elements is called the trace of a matrix which is denoted by $\text{tr}(A)$.

$$\text{tr}(A) = a_{11} + a_{22} + a_{33} + \dots + a_{nn}$$

Remember trace of a matrix is also equal to the sum of eigen value of the matrix. For example:

Properties of Trace of Matrix:

Let A and B be any two square matrices of order n, then

1. $\text{tr}(kA) = k \text{tr}(A)$ where k is a scalar.
2. $\text{tr}(A+B) = \text{tr}(A) + \text{tr}(B)$
3. $\text{tr}(A-B) = \text{tr}(A) - \text{tr}(B)$
4. $\text{tr}(AB) = \text{tr}(BA)$

Determinant of Matrices

Determinant represents the scaling factor of the linear transformation associated with the matrix. For example, in a 2×2 matrix, the determinant represents the area scaling factor.

Properties of Determinant

- $\det(A^T) = \det(A)$
- $\det(AB) = \det(A) \times \det(B)$
- $\det(kA) = k^n \times \det(A)$
- $\det(A) = 0$ implies A is singular.
- If we interchange the row or column in determinant its sign changes from positive to negative.

Rank of a Matrix: Rank of matrix is the number of non-zero rows in the row reduced form or the maximum number of independent rows or the maximum number of independent columns. Rank is denoted as $\text{rank}(A)$ or $\rho(A)$. if A is a non-singular matrix of order n, then rank of A = n i.e. $\rho(A) = n$.

Let A be any $m \times n$ matrix and it has square sub-matrices of different orders. A matrix is said to be of rank r, if it satisfies the following properties:

- It has at least one square sub-matrices of order r who has non-zero determinant.
- All the determinants of square sub-matrices of order $(r+1)$ or higher than r are zero.

Properties of Rank of a Matrix

- If A is a null matrix then $\rho(A) = 0$ i.e. Rank of null matrix is zero.
- If I_n is the $n \times n$ unit matrix then $\rho(A) = n$.
- Rank of a matrix A $m \times n$, $\rho(A) \leq \min(m, n)$. Thus $\rho(A) \leq m$ and $\rho(A) \leq n$.
- $\rho(A_{n \times n}) = n$ if $|A| \neq 0$
- If $\rho(A) = m$ and $\rho(B) = n$ then $\rho(AB) \leq \min(m, n)$.
- If A and B are square matrices of order n then $\rho(AB) \geq \rho(A) + \rho(B) - n$.
- If $A_{m \times 1}$ is a non zero column matrix and $B_{1 \times n}$ is a non zero row matrix then $\rho(AB) = 1$.
- The rank of a skew symmetric matrix cannot be equal to one.

Solution of a System of Linear Equations

Linear equations can have three kind of possible solutions:

- No Solution
- Unique Solution
- Infinite Solution

System of homogeneous linear equations $AX = 0$.

1. $X = 0$. is always a solution; means all the unknowns has same value as zero. (This is also called trivial solution)
2. If $\rho(A) =$ number of unknowns, unique solution.
3. If $\rho(A) <$ number of unknowns, infinite number of solutions.

System of non-homogeneous linear equations $AX = B$.

1. If $\rho[A:B] \neq \rho(A)$, No solution.

2. If $P[A:B] = \rho(A)$ = the number of unknown variables, unique solution.
 3. If $\rho[A:B] = \rho(A) \neq$ number of unknown, infinite number of solutions.
- Here $\rho[A:B]$ is rank of gauss elimination representation of $AX = B$.

There are two states of the Linear equation system:

- **Consistent State:** A System of equations having one or more solutions is called a consistent system of equations.
- **Inconsistent State:** A System of equations having no solutions is called inconsistent system of equations.

Linear dependence and Linear independence of Vector:

- **Linear Dependence:** A set of vectors $X_1, X_2, \dots, X_r$ is said to be linearly dependent if there exist r scalars $k_1, k_2, \dots, k_r$ such that: $k_1 X_1 + k_2 X_2 + \dots + k_r X_r = 0$.
- **Linear Independence:** A set of vectors $X_1, X_2, \dots, X_r$ is said to be linearly independent if for all r scalars $k_1, k_2, \dots, k_r$ such that $k_1 X_1 + k_2 X_2 + \dots + k_r X_r = 0$, then $k_1 = k_2 = \dots = k_r = 0$.

How to Determine Linear Dependency and Independency

Let $X_1, X_2, \dots, X_r$ be the given vectors. Construct a matrix with the given vectors as its rows.

1. If the rank of the matrix of the given vectors is less than the number of vectors, then the vectors are linearly dependent.
2. If the rank of the matrix of the given vectors is equal to the number of vectors, then the vectors are linearly independent.

Eigen Value and Eigen Vector

Eigen vector of a matrix A is a vector represented by a matrix X such that when X is multiplied with matrix A , then the direction of the resultant matrix remains the same as vector X .

Mathematically, above statement can be represented as:

$$AX = \lambda X$$

Where A is any arbitrary matrix, λ are eigen values and X is an eigen vector corresponding to each eigen value.

Here, we can see that AX is parallel to X . So, X is an eigen vector.

How to Find Eigenvalues and Eigen Vectors of Given Matrices

We know that,

$$AX = \lambda X$$

$$AX - \lambda X = 0$$

$$(A - \lambda I) X = 0 \dots (1)$$

Above condition will be true only if $(A - \lambda I)$ is singular. That means,

$$|A - \lambda I| = 0 \dots (2)$$

This is known as characteristic equation of the matrix and the roots of the characteristic equation are the eigen values of the matrix A.

Properties of Eigen Values

- Eigen values of real symmetric and hermitian matrices are real.
- Eigen values of real skew symmetric and skew hermitian matrices are either pure imaginary or zero.
- Eigen values of unitary and orthogonal matrices are of unit modulus $|\lambda| = 1$.
- If $\lambda_1 = \lambda_2 = \dots = \lambda_n$ are the eigen values of A, then $k\lambda_1, k\lambda_2 \dots k\lambda_n$ are eigen values of kA .
- If $\lambda_1, \lambda_2 \dots \lambda_n$ are the eigen values of A, then $1/\lambda_1, 1/\lambda_2 \dots 1/\lambda_n$ are eigen values of A^{-1} .
- If $\lambda_1, \lambda_2 \dots \lambda_n$ are the eigen values of A, then $\lambda_1^k, \lambda_2^k \dots \lambda_n^k$ are eigen values of A^k .
- Eigen values of A = Eigen Values of A^T (Transpose).
- Sum of Eigen Values = Trace of A (Sum of diagonal elements of A).
- Product of Eigen Values = $|A|$.
- Maximum number of distinct eigen values of A = Size of A.
- If A and B are two matrices of same order then, Eigen values of $AB =$ Eigen values of BA .

LU Decomposition

LU Decomposition (also known as LU Factorization) is a method used to solve a system of linear equations. It decomposes a given square matrix **A** into the product of two matrices **L** (Lower Triangular Matrix) and **U** (Upper Triangular Matrix), such that:
 $A = L \cdot U$

Where:

- L is a lower **triangular matrix** with ones on the diagonal.
- U is an upper triangular matrix.